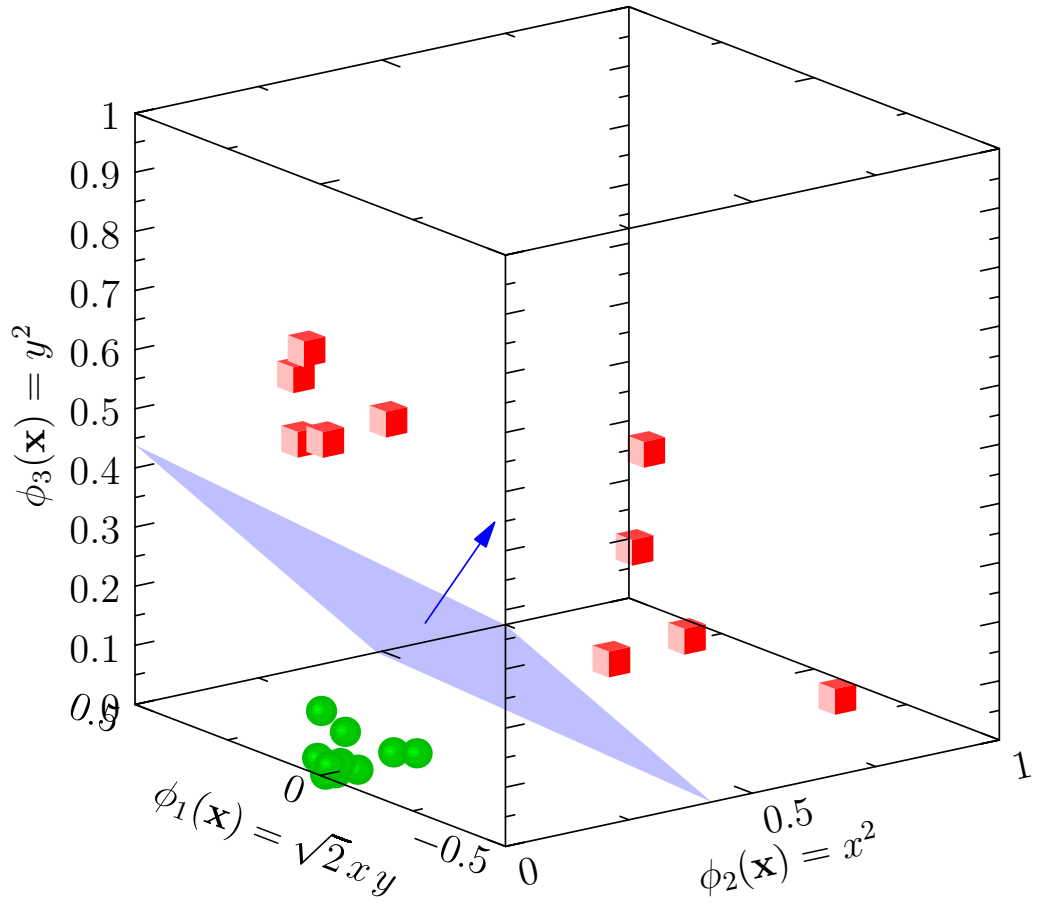
Mathematics of Machine Learning

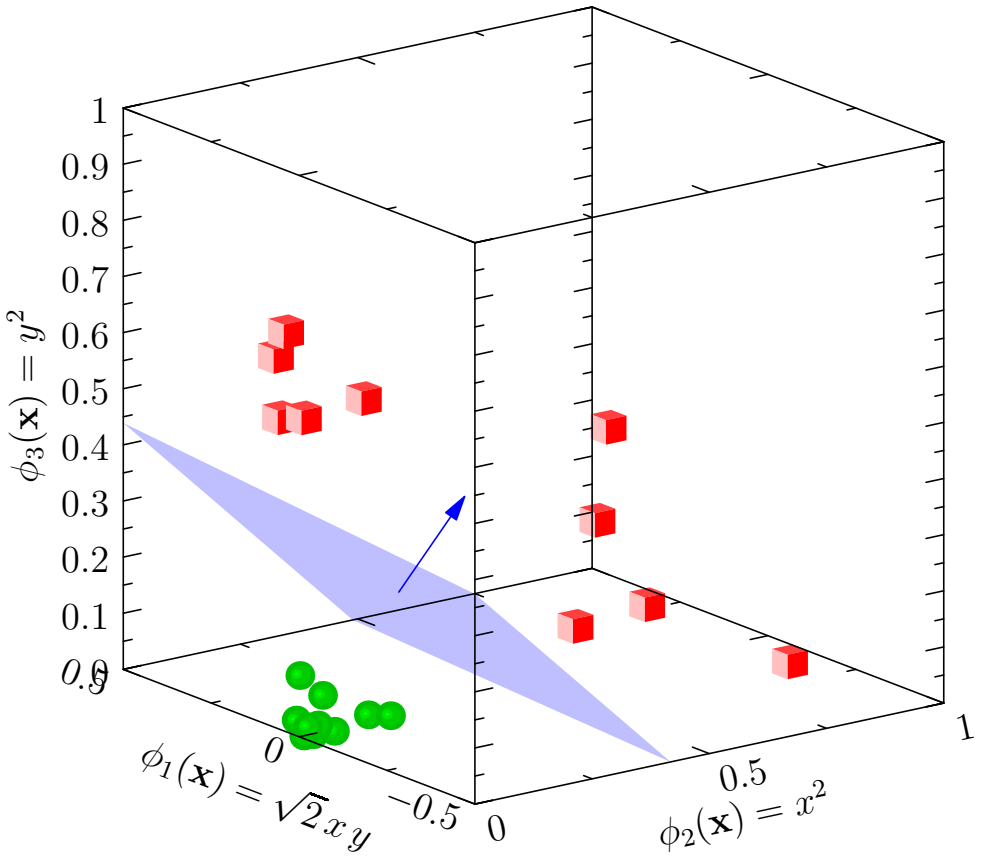
Kernel Trick



The Kernel Trick, SVMs, Regression

Outline

1. **Recap: the Kernel Trick**
2. Mercer's Theorem
3. Constructing Kernels
4. Kernels for Structured Data
5. Beyond Classification



Recap: the Kernel Trick

- SVM kernels are functions of two variables that factorise

$$K(\boldsymbol{x},\boldsymbol{y}) = \langle \boldsymbol{\phi}(\boldsymbol{x}), \boldsymbol{\phi}(\boldsymbol{y}) \rangle = \sum_i \phi_i(\boldsymbol{x})\phi_i(\boldsymbol{y})$$

  with real-valued $\phi_i(\boldsymbol{x})$

- Such a K is positive semi-definite and, conversely, any positive semi-definite K factorises (this lecture)
- The SVM dual problem only needs the inner products

$$\max_{\boldsymbol{\alpha}} \sum_{k=1}^{m} \alpha_k - \frac{1}{2} \sum_{k,l=1}^{m} \alpha_k \alpha_l y_k y_l K(\boldsymbol{x}_k, \boldsymbol{x}_l)$$

  subject to $\sum_{k=1}^{m} y_k \alpha_k = 0$ and $0 \leq \alpha_k \leq C$

- This is the **kernel trick**: we never compute $\boldsymbol{\phi}(\boldsymbol{x})$

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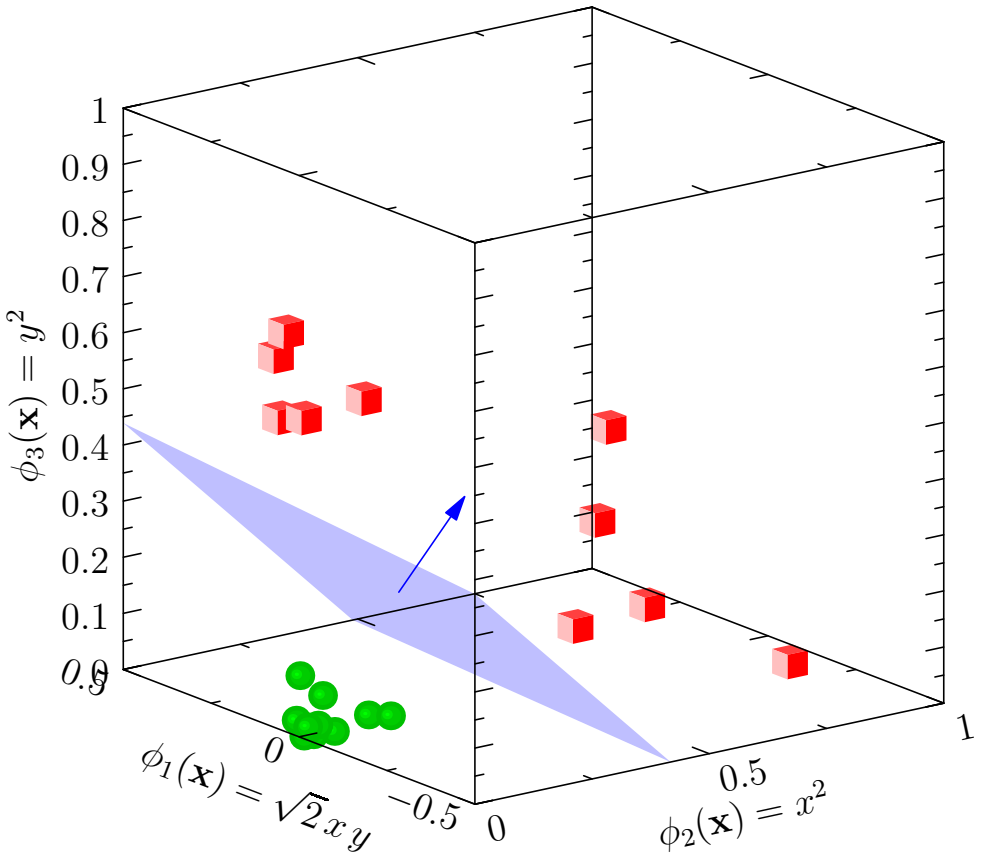
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Recap on Eigen Systems

- Recall for a symmetric $(n \times n)$ matrix $\mathbf{M}$ an eigenvector, $\boldsymbol{v}$

$$\mathbf{M}\boldsymbol{v} = \lambda \boldsymbol{v}$$

- There are n independent eigenvectors $\boldsymbol{v}^{(i)}$ with real eigenvalues $\lambda^{(i)}$
- The eigenvectors can be chosen orthonormal: $\boldsymbol{v}^{(i)\mathsf{T}}\boldsymbol{v}^{(j)} = 0$ if $i \neq j$ $(= 1$ if $i = j)$
- Forming a matrix of eigenvectors $\mathbf{V} = (\boldsymbol{v}^{(1)}, \boldsymbol{v}^{(2)}, \ldots \boldsymbol{v}^{(n)})$ the matrix satisfies

$$\mathbf{V}^{\mathsf{T}}\mathbf{V} = \mathbf{V}\mathbf{V}^{\mathsf{T}} = \mathbf{I}$$

- Such matrices are said to be orthogonal

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Eigen Decomposition

- From the eigenvalue equation $\boldsymbol{M}\boldsymbol{v}^{(k)} = \lambda^{(k)}\boldsymbol{v}^{(k)}$

$$\boldsymbol{M}\boldsymbol{V} = \boldsymbol{V}\boldsymbol{\Lambda} \qquad \text{where} \quad \boldsymbol{\Lambda} = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix}$$

- Multiplying on the right by $\boldsymbol{V}^{\mathsf{T}}$ we get

$$\boldsymbol{M} = \boldsymbol{V}\boldsymbol{\Lambda}\boldsymbol{V}^{\mathsf{T}} = \sum_{k=1}^{n} \lambda^{(k)} \boldsymbol{v}^{(k)} \boldsymbol{v}^{(k)\mathsf{T}}$$

Or

$$M_{ij} = \sum_{k=1}^{n} \lambda^{(k)} v_i^{(k)} v_j^{(k)}$$

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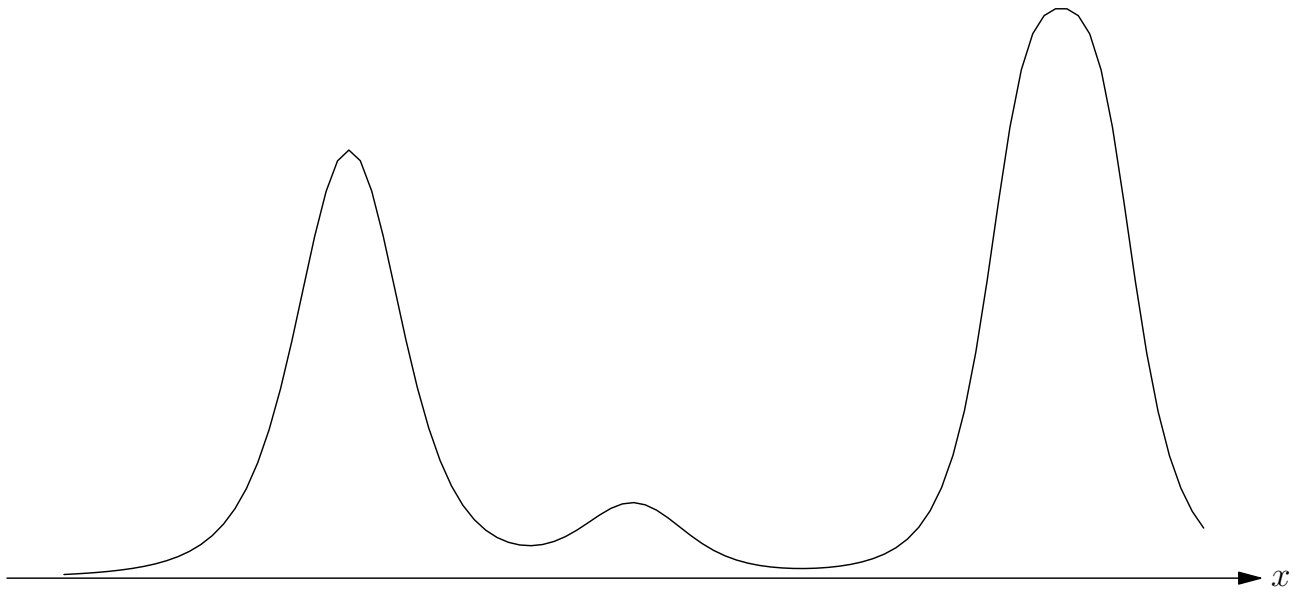
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$$M_{ij} = \sum_{k=1}^{n} \lambda^{(k)} v_i^{(k)} v_j^{(k)} = \sum_{k=1}^{n} u_i^{(k)} u_j^{(k)} = \langle \boldsymbol{u}_i, \boldsymbol{u}_j \rangle$$

$u_i^{(k)} = \sqrt{\lambda^{(k)}} v_i^{(k)}$ (real only if $\lambda^{(k)} \geq 0$)

Limit Process

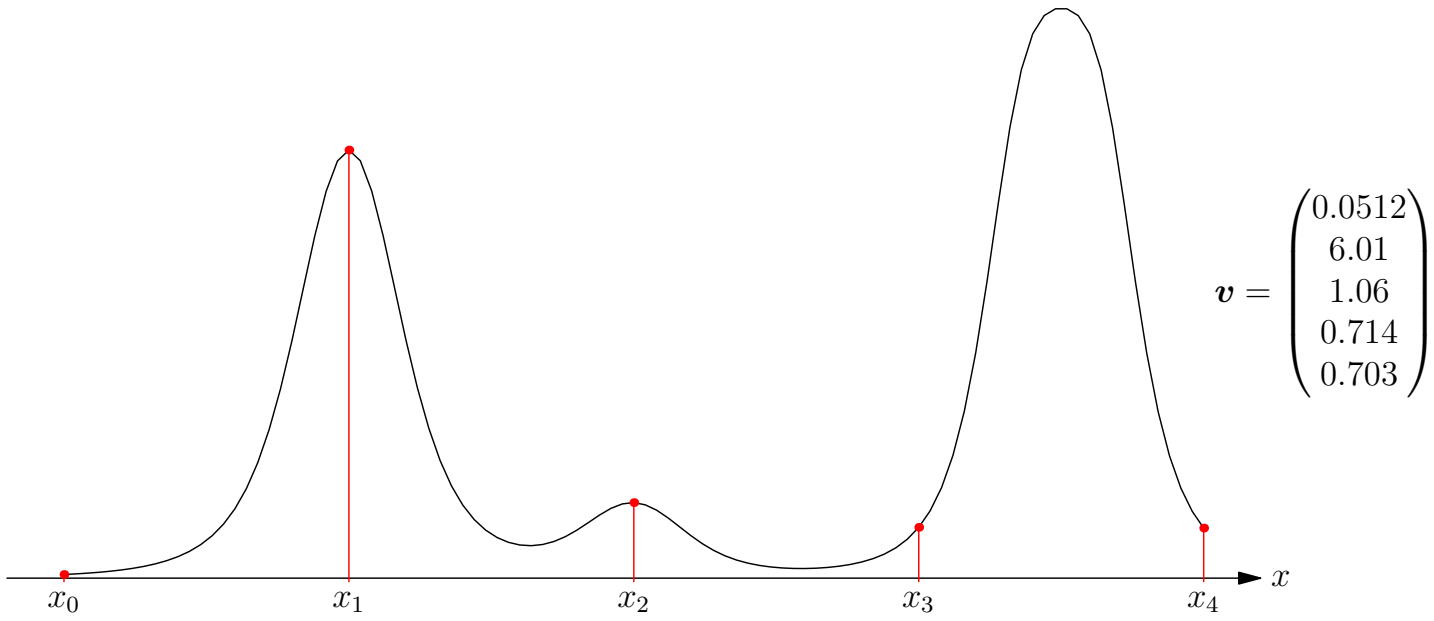
- Consider sampling a function at a set of points



- In the limit where the number of sample points goes to infinity the vector more closely approximates a function
- The kth component of the vector corresponds to $f(x_k)$

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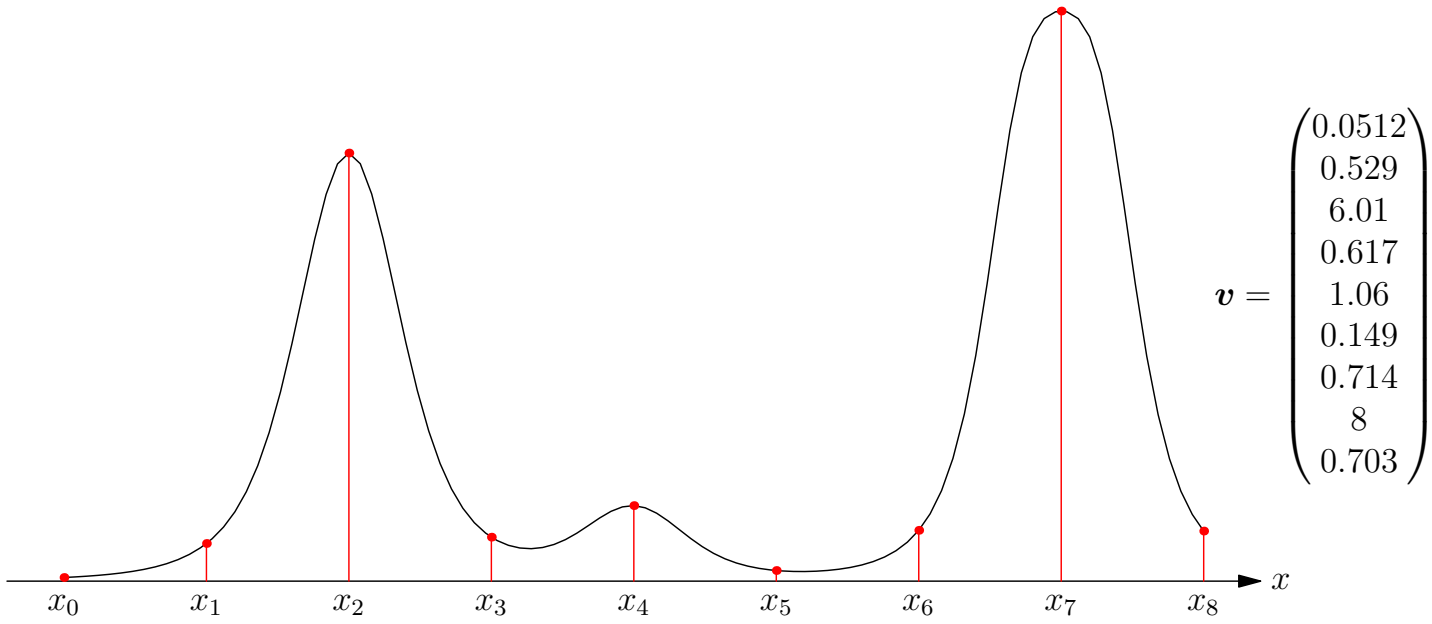


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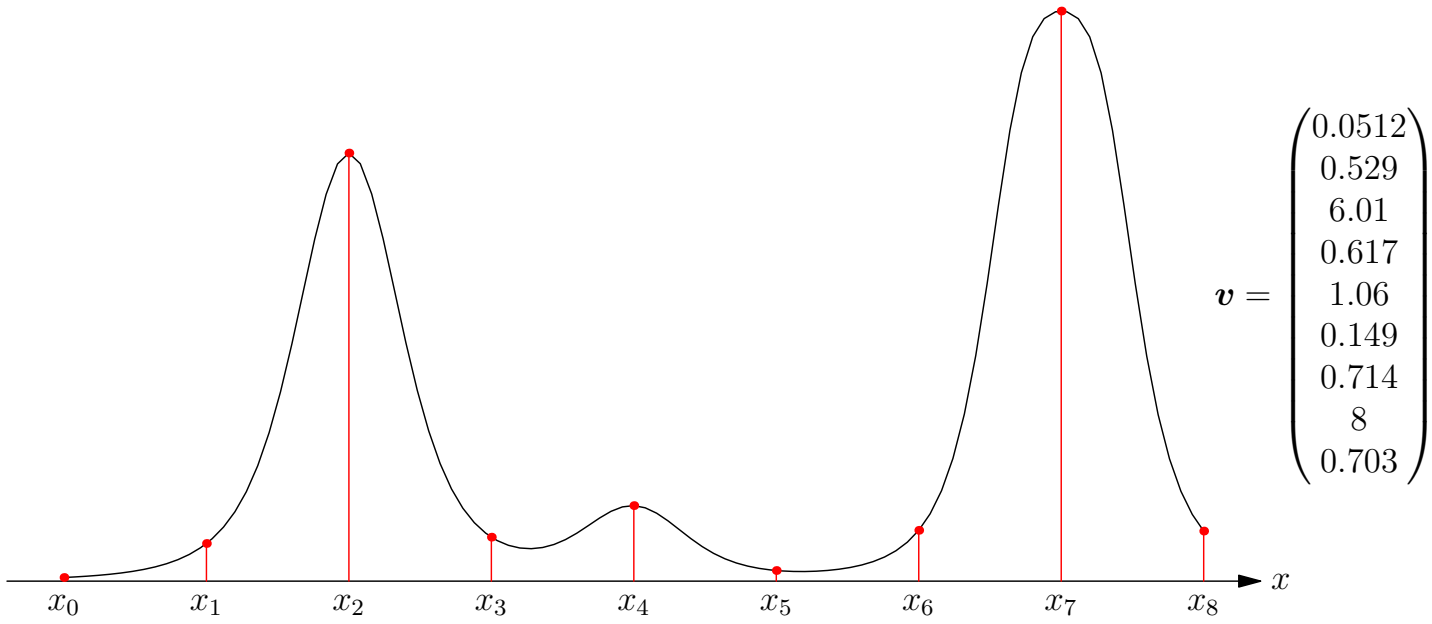


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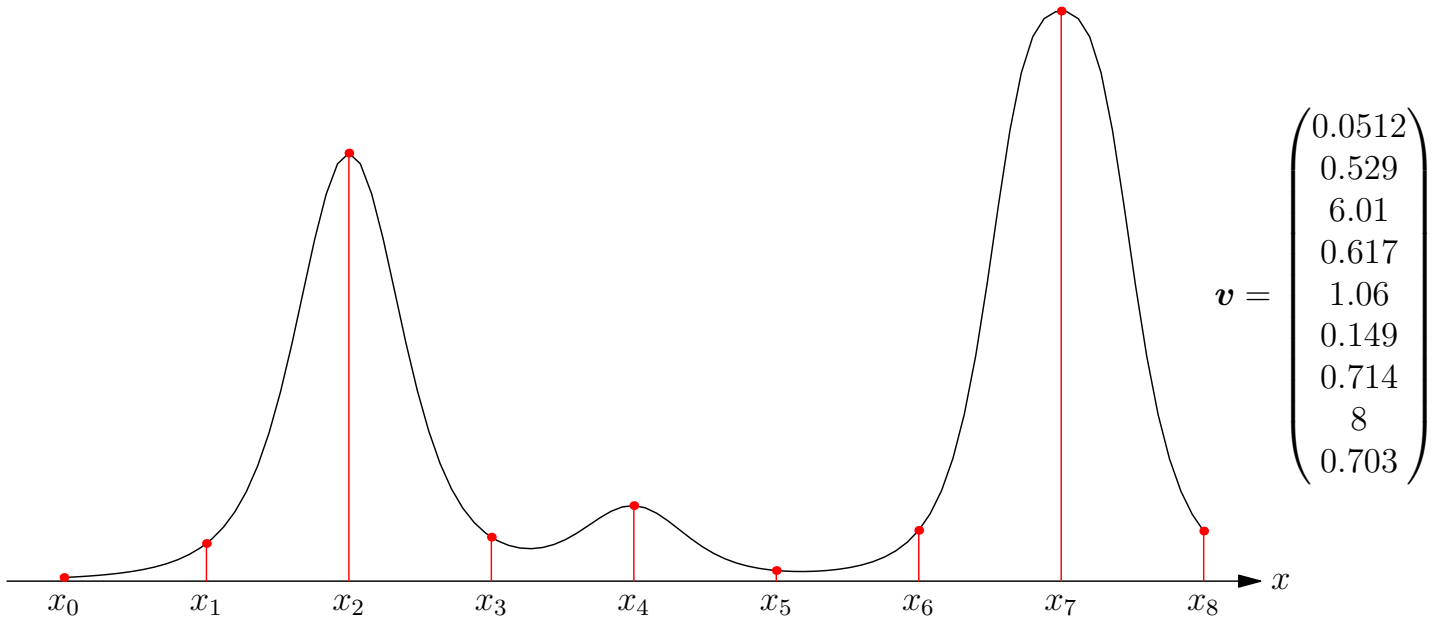


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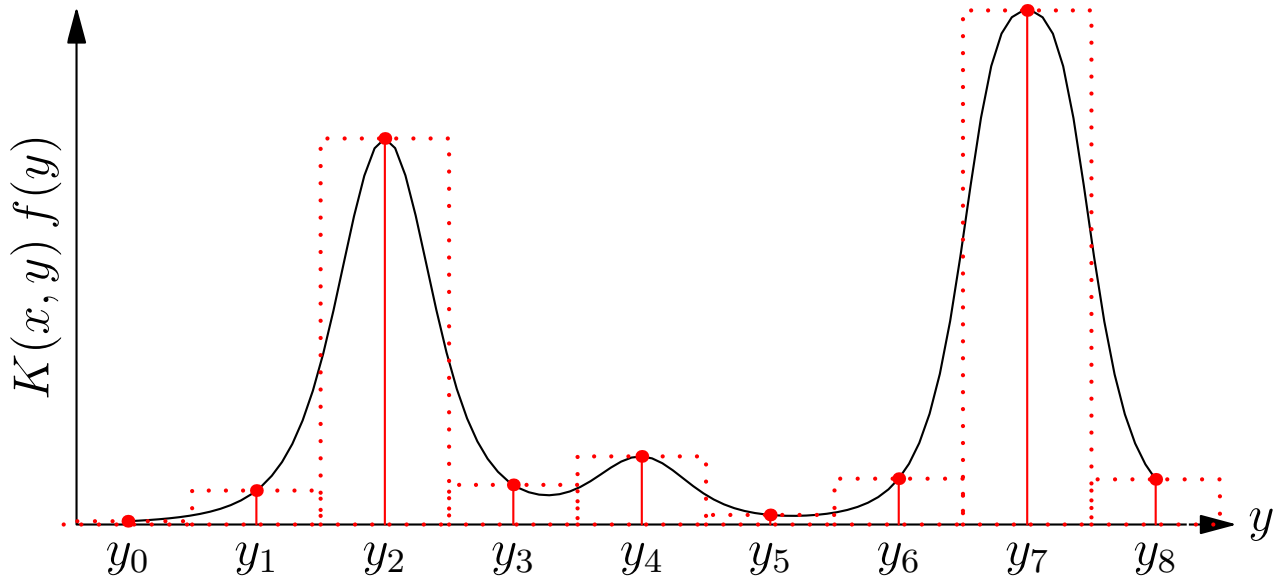
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Linear Operators

- Recall a linear operator on functions, $\mathcal{T}[f(x)]$, can often be represented by a kernel

$$\mathcal{T}[f(x)] = \int_{y\in\mathcal{I}} K(x,y)\,f(y)\,\mathrm{d}y$$

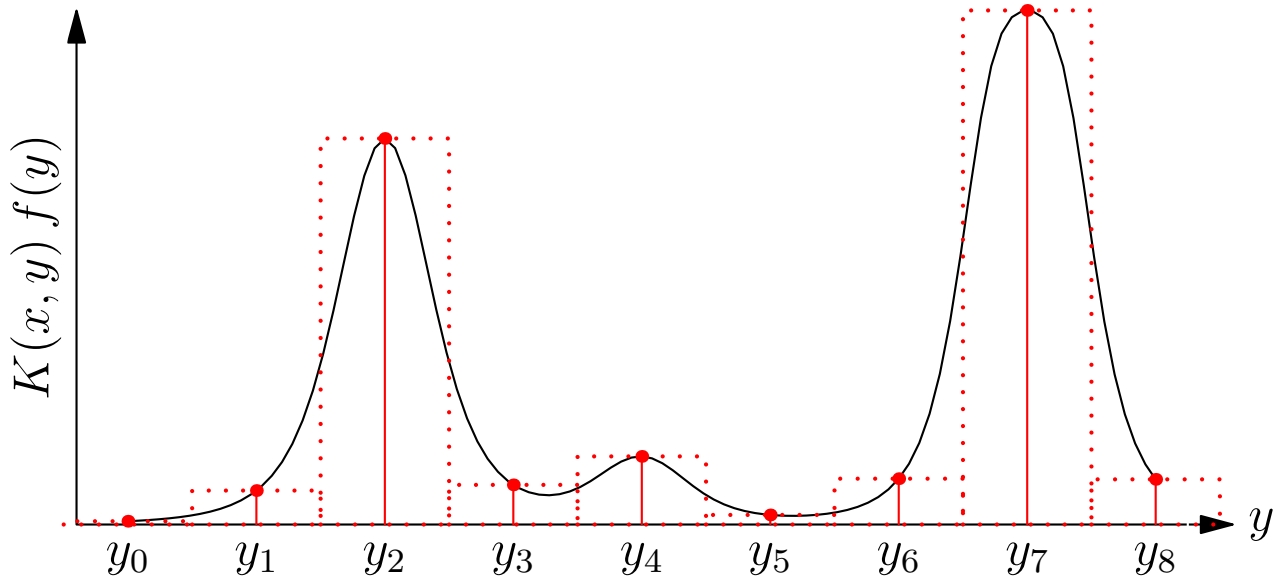


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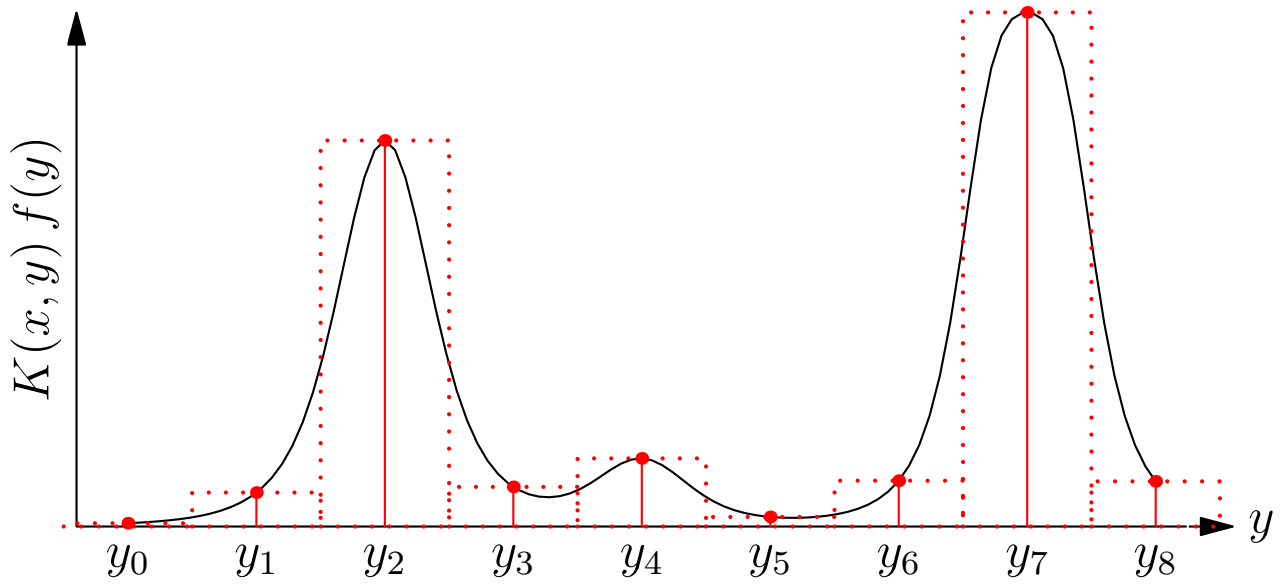


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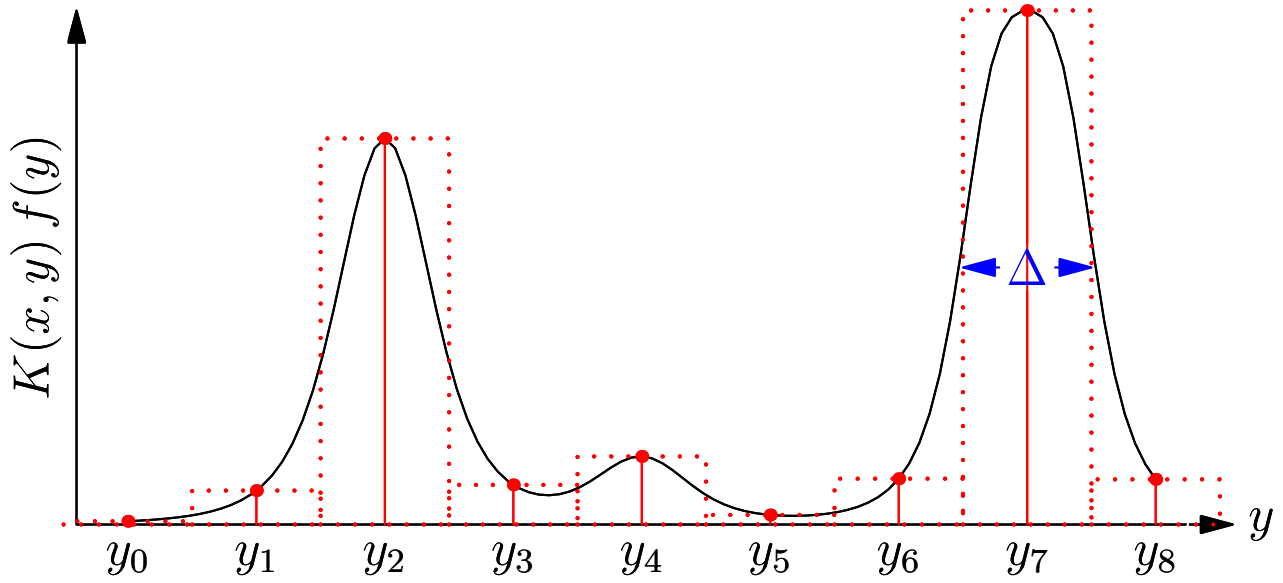


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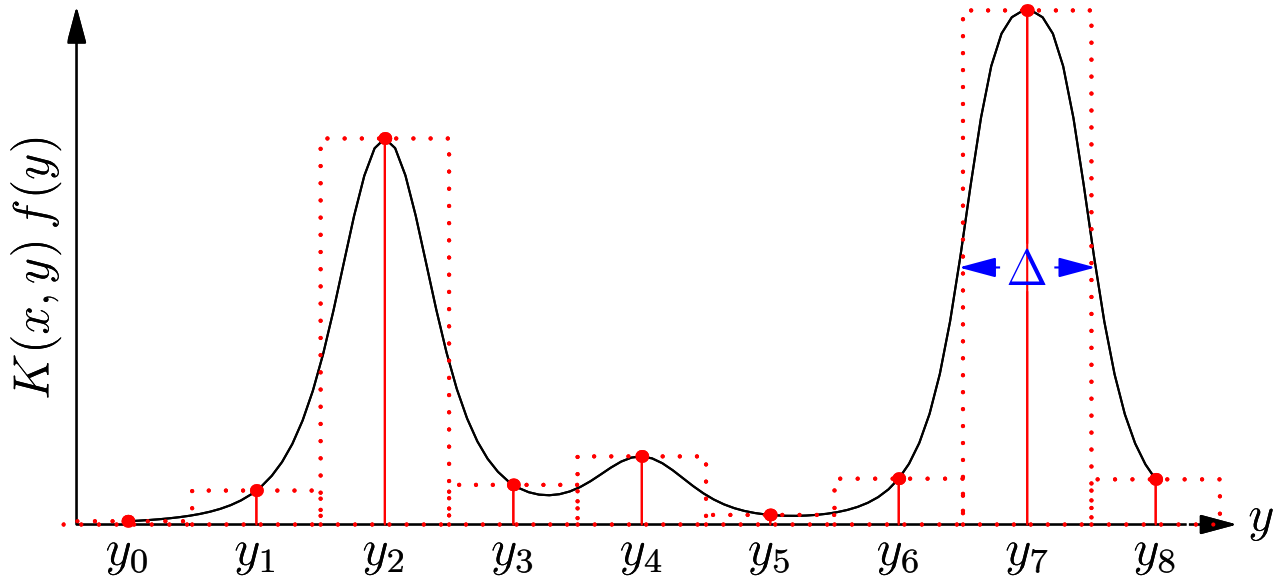


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- By analogy for a symmetric function of two variables we can define an *eigenfunction*

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- In general there will be a denumerable set of eigenfunctions $\psi^{(k)}(\boldsymbol{x})$ where

$$K(\boldsymbol{x},\boldsymbol{y}) = \sum_k \lambda^{(k)}\psi^{(k)}(\boldsymbol{x})\psi^{(k)}(\boldsymbol{y})$$

- For a positive semi-definite kernel (on a bounded domain) this is **Mercer's theorem**

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- Note that for $\phi^{(k)}(\boldsymbol{x})$ to be real $\lambda^{(k)} \geq 0$ for all k
- If $\lambda^{(k)} < 0$ then $\langle \boldsymbol{\phi}(\boldsymbol{x}), \boldsymbol{\phi}(\boldsymbol{x}) \rangle = \|\phi(\boldsymbol{x})\|^2$ might be negative and "distance" between points in the extended feature space can be negative!
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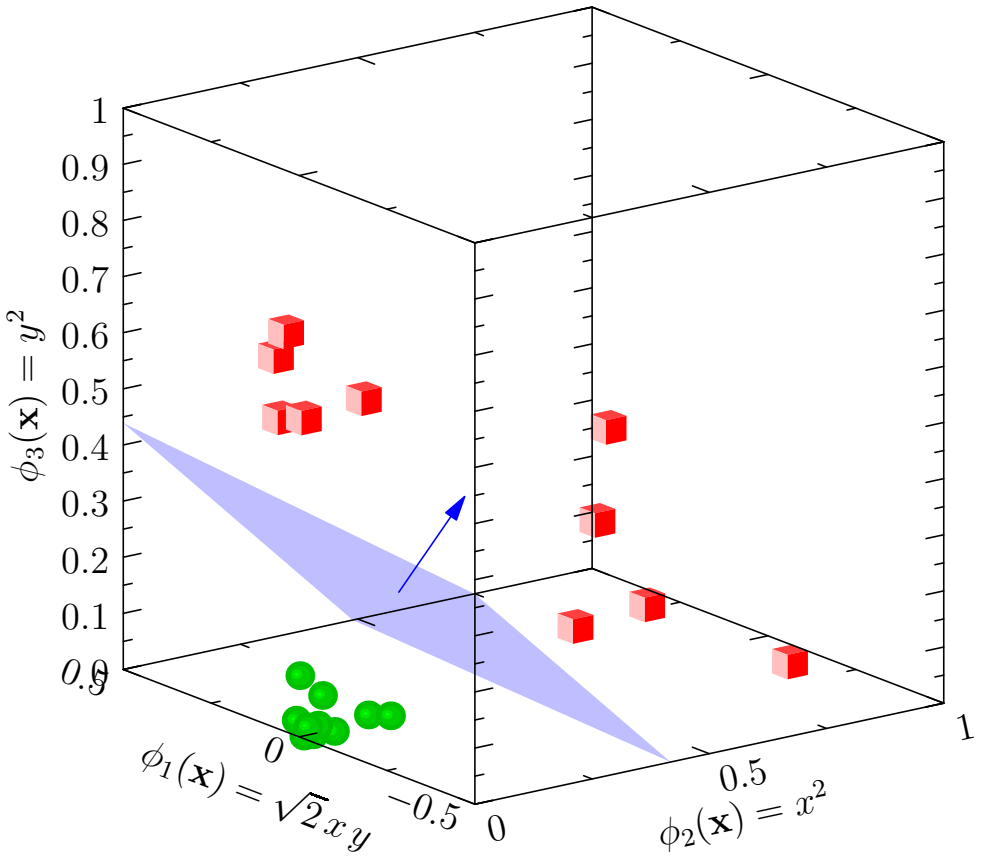
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5. Beyond Classification



Positive Semi-Definite Kernels

- Kernels (or matrices) that have eigenvalues $\lambda^{(k)} \geq 0$ are called positive semi-definite
- (If the eigenvalues are strictly positive $\lambda^{(k)} > 0$ the kernels or matrices are called positive definite)
- Positive semi-definite kernels can always be decomposed into a sum of real functions

$$K(\boldsymbol{x},\boldsymbol{y}) = \langle \boldsymbol{\phi}(\boldsymbol{x}), \boldsymbol{\phi}(\boldsymbol{y}) \rangle$$

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Properties of Positive Semi-Definiteness

- Since

$$K(\boldsymbol{x},\boldsymbol{y}) = \left\langle \boldsymbol{\phi}(\boldsymbol{x}),\boldsymbol{\phi}(\boldsymbol{y}) \right\rangle$$

- An immediate consequence is that for any function $f(\boldsymbol{x})$

$$\int f(\boldsymbol{x}) K(\boldsymbol{x},\boldsymbol{y}) f(\boldsymbol{y}) \mathrm{d}\boldsymbol{x} \mathrm{d}\boldsymbol{y} = \int f(\boldsymbol{x}) \left\langle \boldsymbol{\phi}(\boldsymbol{x}),\boldsymbol{\phi}(\boldsymbol{y}) \right\rangle f(\boldsymbol{y}) \mathrm{d}\boldsymbol{x} \mathrm{d}\boldsymbol{y}$$

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$$= \left\langle \int f(\boldsymbol{x}) \boldsymbol{\phi}(\boldsymbol{x}) \mathrm{d}\boldsymbol{x}, \int f(\boldsymbol{y}) \boldsymbol{\phi}(\boldsymbol{y}) \mathrm{d}\boldsymbol{y} \right\rangle$$

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- The following statements are equivalent
  - ⋆ $K(\boldsymbol{x},\boldsymbol{y})$ is positive semi-definite (written $K(\boldsymbol{x},\boldsymbol{y}) \succeq 0$)
  - ⋆ The eigenvalues of $K(\boldsymbol{x},\boldsymbol{y})$ are non-negative
  - ⋆ The kernel can be written

$$K(\boldsymbol{x},\boldsymbol{y}) = \left\langle \boldsymbol{\phi}(\boldsymbol{x}),\boldsymbol{\phi}(\boldsymbol{y}) \right\rangle$$

    where the $\phi^{(k)}(\boldsymbol{x})$'s are real functions
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Adding Kernels

- We can construct SVM kernels from other kernels

- If $K_1(\boldsymbol{x},\boldsymbol{y})$ and $K_2(\boldsymbol{x},\boldsymbol{y})$ are valid kernels then so is $K_3(\boldsymbol{x},\boldsymbol{y}) = K_1(\boldsymbol{x},\boldsymbol{y}) + K_2(\boldsymbol{x},\boldsymbol{y})$

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Q &= \int f(\boldsymbol{x}) K_3(\boldsymbol{x},\boldsymbol{y}) f(\boldsymbol{y}) \mathrm{d}\boldsymbol{x} \mathrm{d}\boldsymbol{y} \\
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&= \int f(\boldsymbol{x}) K_1(\boldsymbol{x},\boldsymbol{y}) f(\boldsymbol{y}) \mathrm{d}\boldsymbol{x} \mathrm{d}\boldsymbol{y} + \int f(\boldsymbol{x}) K_2(\boldsymbol{x},\boldsymbol{y}) f(\boldsymbol{y}) \mathrm{d}\boldsymbol{x} \mathrm{d}\boldsymbol{y} \geq 0
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Product of Kernels

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$$K_1(\boldsymbol{x},\boldsymbol{y}) = \sum_i \phi_i^{(1)}(\boldsymbol{x})\phi_i^{(1)}(\boldsymbol{y}), \qquad K_2(\boldsymbol{x},\boldsymbol{y}) = \sum_j \phi_j^{(2)}(\boldsymbol{x})\phi_j^{(2)}(\boldsymbol{y})$$

then

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Exponentiating Kernels

- If $K(\boldsymbol{x},\boldsymbol{y})$ is a valid kernel so is $K^n(\boldsymbol{x},\boldsymbol{y})$ (by induction)
  - ⋆ Assume $K(\boldsymbol{x},\boldsymbol{y}) \succeq 0$ this satisfies base case
  - ⋆ If $K^{n-1}(\boldsymbol{x},\boldsymbol{y}) \succeq 0$ then

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- and $\exp(K(\boldsymbol{x},\boldsymbol{y}))$ is also a valid kernel since

$$\mathrm{e}^{K(\boldsymbol{x},\boldsymbol{y})} = \sum_{i=0}^{\infty} \frac{1}{i!} K^i(\boldsymbol{x},\boldsymbol{y}) = 1 + K(\boldsymbol{x},\boldsymbol{y}) + \frac{1}{2}K^2(\boldsymbol{x},\boldsymbol{y}) + \cdots$$

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Polynomial Kernels

- A constant $c \geq 0$ is a kernel: $c = \sqrt{c}\sqrt{c}$, so $\phi(\boldsymbol{x}) = \sqrt{c}$
- $\boldsymbol{x}^\mathsf{T}\boldsymbol{y}$ is a kernel (with $\boldsymbol{\phi}(\boldsymbol{x}) = \boldsymbol{x}$), so the sum $c + \boldsymbol{x}^\mathsf{T}\boldsymbol{y}$ is too
- Hence the **polynomial kernel**

$$K(\boldsymbol{x}, \boldsymbol{y}) = \left(c + \boldsymbol{x}^\mathsf{T}\boldsymbol{y}\right)^d$$

  is a kernel for $d = 1, 2, \ldots$

- Its features $\boldsymbol{\phi}(\boldsymbol{x})$ are all the monomials of the x_i of degree up to d (suitably weighted)
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RBF Kernel

- Now $\boldsymbol{x}^\mathsf{T}\boldsymbol{y} = \langle \boldsymbol{x}, \boldsymbol{y} \rangle$ is a valid kernel because it is an inner product of functions $\boldsymbol{\phi}(\boldsymbol{x}) = \boldsymbol{x}$
- For $\gamma > 0$ we have $2\gamma \boldsymbol{x}^\mathsf{T}\boldsymbol{y} \succeq 0$
- Thus $\exp(2\gamma \boldsymbol{x}^\mathsf{T}\boldsymbol{y}) \succeq 0$
- If $K(\boldsymbol{x}, \boldsymbol{y}) \succeq 0$ then $g(\boldsymbol{x})K(\boldsymbol{x}, \boldsymbol{y})g(\boldsymbol{y}) \succeq 0$

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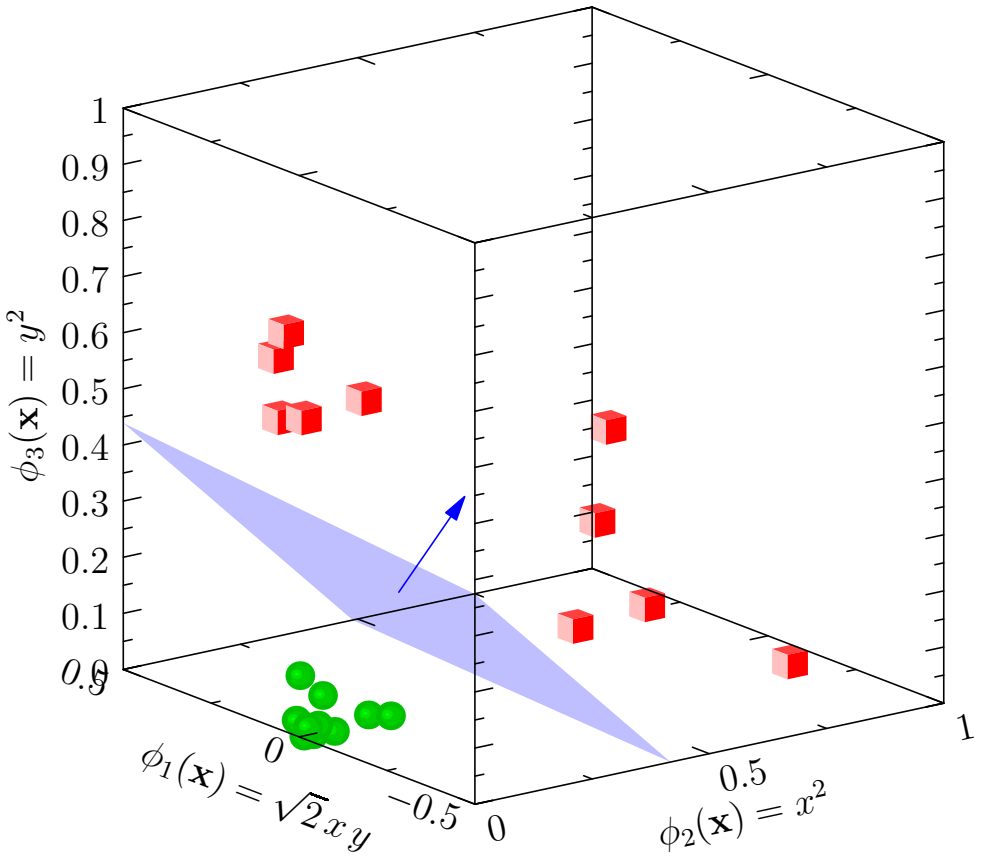
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Outline

1. Recap: the Kernel Trick
2. Mercer's Theorem
3. Constructing Kernels
4. **Kernels for Structured Data**
5. Beyond Classification



Other Kernels

- The success of SVMs has meant that researchers try to increase the area of application
- The condition that an SVM kernel must be positive semi-definite is quite restrictive
- The key to finding new kernels is to use the properties of kernels to build more complicated kernels from simpler ones
- There has been a cottage industry of researchers finding smart kernels for complicated data
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Spectrum Kernel

- A simple way to compare documents is to collect a histogram of all occurrences of substrings of length ℓ
- This is known as an ℓ-spectrum
- The ℓ-spectrum kernel is the inner product of the two histograms

$$s = \texttt{statistics} \quad \mathcal{S}_3(s) = \{\texttt{sta},\texttt{tat},\texttt{ati},\texttt{tis},\texttt{ist},\texttt{sti},\texttt{tic},\texttt{ics}\}$$
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- Using clever dynamic-programming techniques this can be done relatively efficiently
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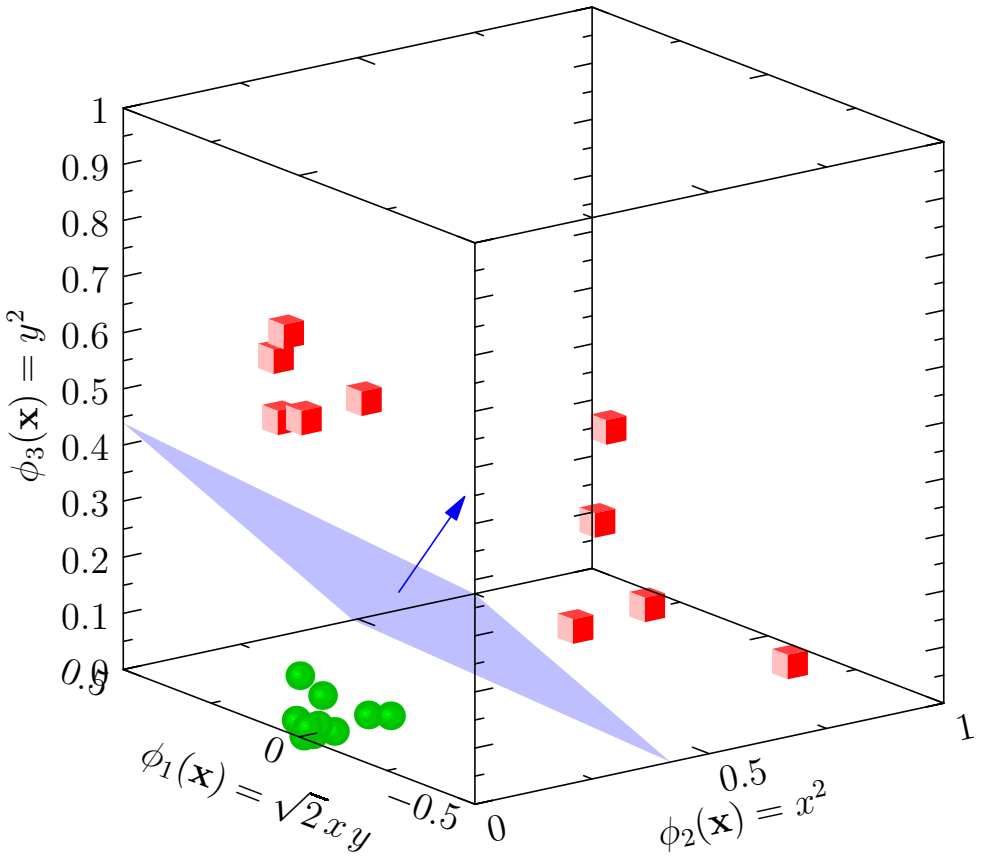
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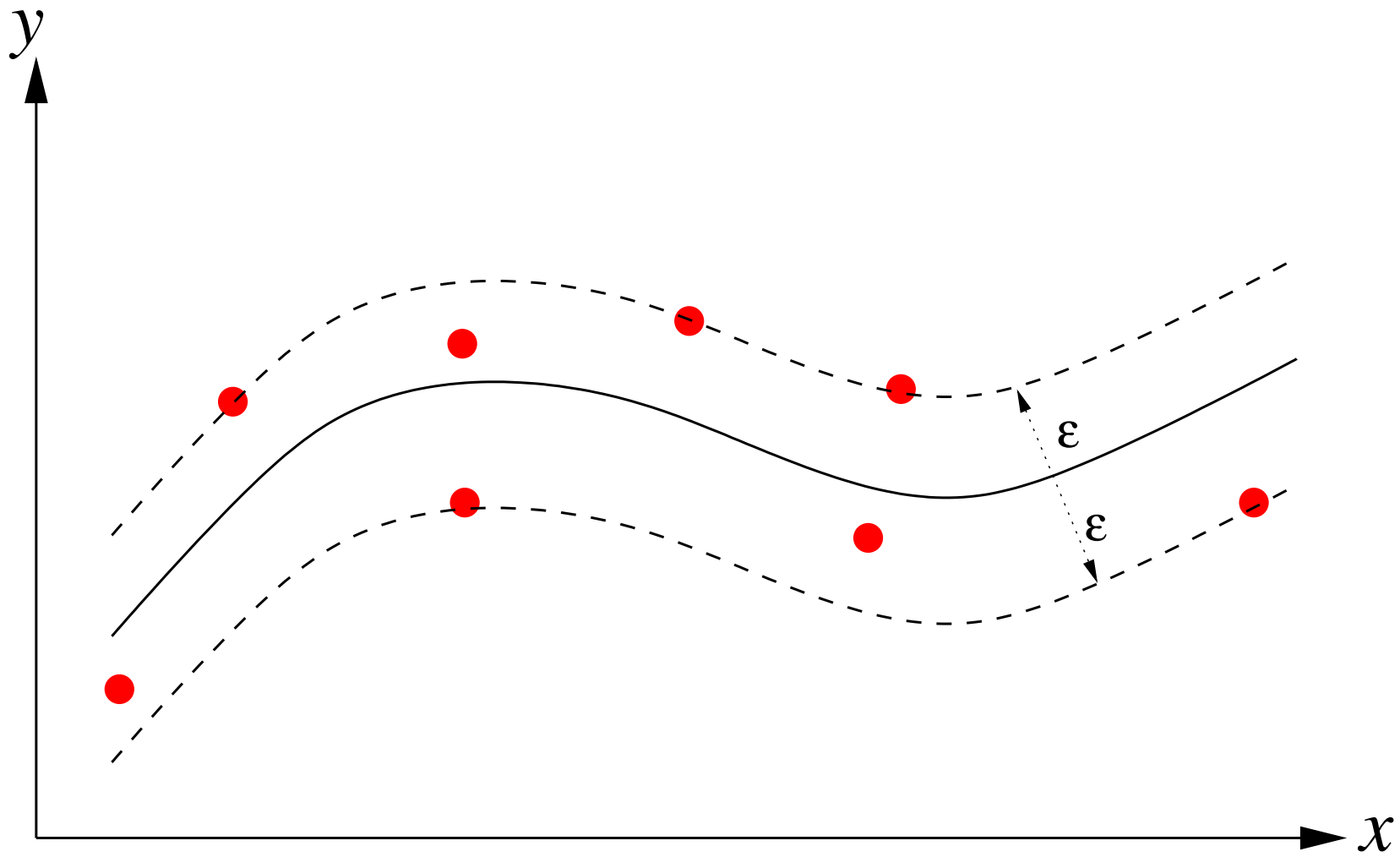
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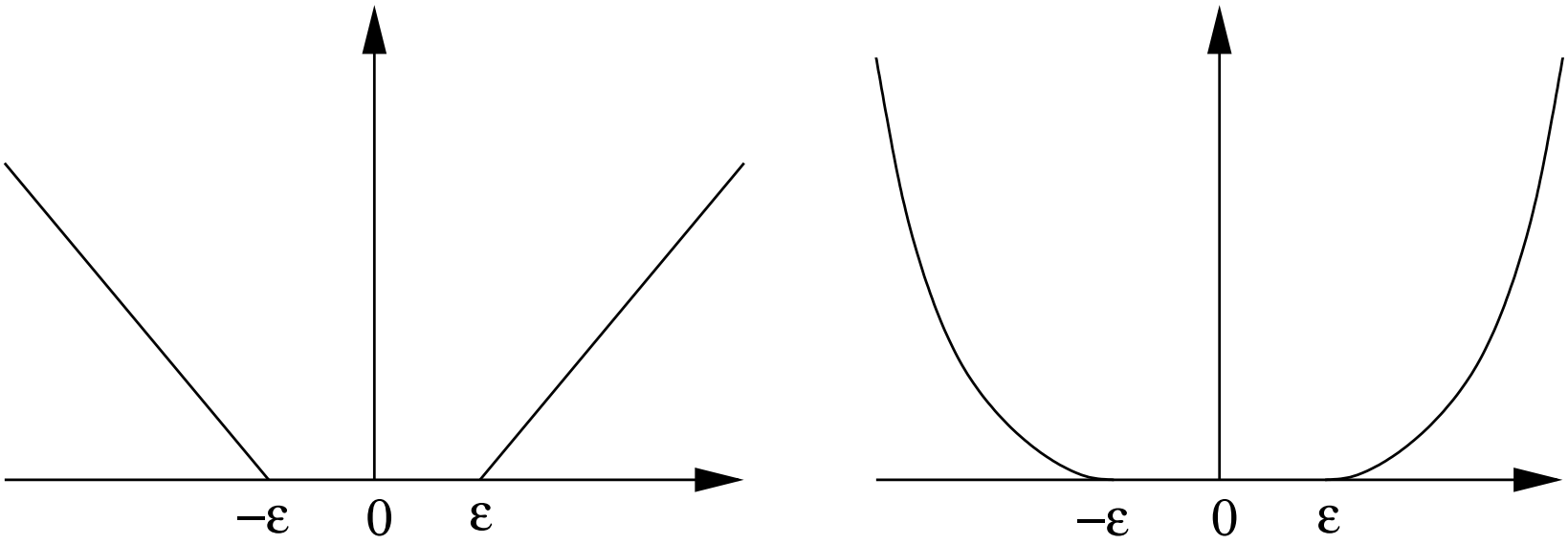
Regression with Margins

- SVMs can be modified to perform regression



Error Functions

- Can introduce slack variables with different errors



- This can be transformed to a quadratic programming problem

Ridge Regression Using Kernels

- We can also solve regression problems without using margins
- Kernel ridge regression minimises

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  where $\eta \|\boldsymbol{w}\|^2$ is a regularisation term

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